

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (EC) Examination

April 2013

EC 208 Analog Electronic Circuits

Date: 25.04.2013, Thursday

Time: 01:30 pm To 04:30 pm

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) In regard to time base generators define following terms: (i) Sweep time (ii) [03]
Restoration Time (iii) Slope Error

(b) Derive the equation of efficiency of class A power Amplifier [04]

Q - 2 (a) Explain Barkhausen Criterion for Oscillation. With neat illustration describe Wien [07]
Bridge Oscillator.

(b) With the help of neat circuit diagram and waveforms explain the working of UJT as a [07]
relaxation oscillator.

OR

Q - 2 (a) Explain the piezoelectric effect. With neat sketch of equivalent circuit explain [07]
Crystal Oscillator.

(b) Describe quiescent conditions and formation of sweep in bootstrap time base [07]
generator with necessary waveforms.

Q - 3 (a) Describe simple current sweep circuit and waveform for inductor current. [04]

(b) In the UJT sweep circuit $V_{BB} = 20\text{ V}$, $V_{YY} = 50\text{ V}$, $R = 5\text{ k}\Omega$, $R_{B1} = R_{B2} = 0\Omega$ and [04]
 $C = 0.01\mu\text{F}$. UJT fires when $V_c = 10.6\text{ V}$ and goes to OFF state when $V_c = 2.8\text{ V}$,
Calculate (i) sweep signal amplitude (ii) sweep speed error (iii) displacement error
(iv) sweep duration.

(c) Draw the block diagram of successive – approximation type Analog to Digital [06]
Converter.

OR

Q - 3 (a) Describe linearity correction in current sweep circuit through adjustment of driving [04]
waveform.

(b) In transistor current time-base generator, $V_{cc} = 20\text{ V}$, $L = 200\text{ mH}$, yoke resistance R_L [04]
 $= 20\text{ ohms}$, $R_{CS} = 5\text{ ohms}$. For a $500\mu\text{s}$ sweep, draw waveforms of inductor current
(i_L) and V_{CE} indicating voltage levels and time constants.

(c) Draw the block diagram of flash type Analog to Digital Circuits. [06]

SECTION – II

- Q - 4 (a)** Calculate the Output voltage V_o and Zener current in the Regulator circuit of Figure 1 for $R_L = 1\text{k}\Omega$ [04]

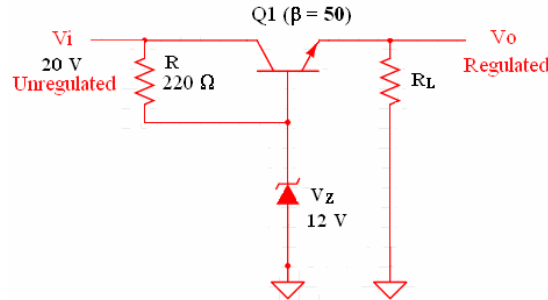


Figure 1: Regulator Circuit

- (b)** Draw output V_o of Series clipper circuit of Figure 2. [04]

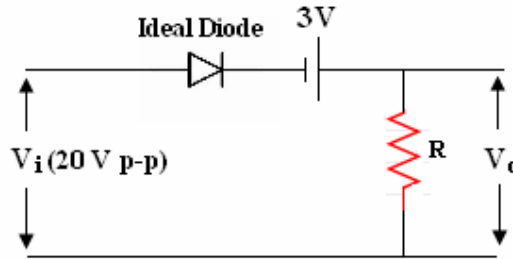


Figure 2: Series Clipper Circuit

- (c)** What is the function of differential amplifier? Explain it with linear active model. [05]

OR

- Q - 4 (a)** Explain series voltage regulator with necessary block diagram [04]
- (b)** Draw output V_o of Series clipper circuit of Figure 3. [04]

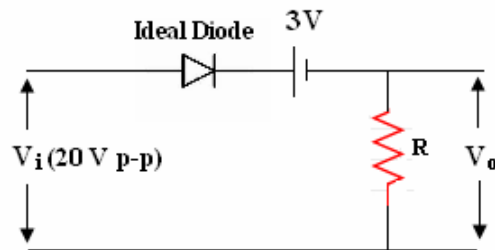


Figure 3: Series Clipper Circuit

- (c)** What is CMRR? Derive its equation. [05]

- Q - 5 (a)** What is the relationship between the transfer gain with feedback A_f and that without feedback A ? Define (a) Positive feedback (b) Negative feedback. In addition, enlist five characteristics of an amplifier which are modified by negative feedback. [07]
- (b)** With the help of the $V - I$ characteristic and load line, explain (i) the monostable and [07]
(ii) the bistable operations of current controlled negative resistance devices

OR

- Q - 5 (a)** What is the mean of a current controllable negative resistance device and voltage controlled negative resistance device. Give examples of both type of negative resistance device. [04]
- (b)** Draw the output of Clamper circuit of Figure 4. [04]

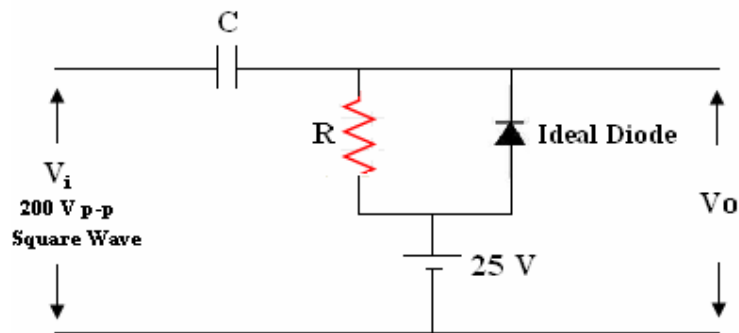


Figure 4: Clamper Circuit

- (c)** Explain the operation of Tunnel Diode illustrating its $V-I$ Characteristics [06]
- Q - 6 Attempt Any Two** [08]
- (a)** Describe $R-2R$ Ladder type Digital to Analog conversion with necessary calculations.
- (b)** Consider the situation, where the first set of signals is $V_1 = +50\mu V$ and $V_2 = -50\mu V$ and the second set of the signals is $V_1 = 1050\mu V$ and $V_2 = 950\mu V$. If the CMRR is 100, calculate the percentage difference in output voltage obtained for the two sets of input signals. Repeat part a if CMRR 10,000.
- (c)** With necessary illustrations describe transformer coupled Class-A power amplifier
- (d)** Describe Transistor Clipper with neat illustrations
